Input and Output Devices in Detail

An **input device** sends data and instructions *into* the computer. An **output device** presents the processed result *from* the computer to the user. Together they form the link between the user and the machine.

Data flow: Input device → CPU (processing) → Output device

Input Devices

Definition

An input device converts human actions (typing, speaking, touching, scanning) into **digital signals (0s and 1s)** that the computer can process.

Characteristics

- Only sends data to the computer (one-way)
- Converts data into machine-readable form
- Connects through USB, Bluetooth, PS/2 or built-in ports

Classification of Input Devices

1. Keyboard

The main device for entering text, numbers and commands.

Key groups:

- **Alphanumeric keys:** letters and numbers (QWERTY layout)
- **Function keys (F1-F12):** shortcuts such as F5 (refresh)
- **Numeric keypad:** quick number entry
- **Navigation keys:** arrows, Home, End, Page Up/Down
- **Modifier keys:** Shift, Ctrl, Alt
- **Special keys:** Enter, Esc, Backspace, Delete, Tab, Caps Lock

Types: wired, wireless, mechanical, membrane, virtual (on-screen), ergonomic

2. Pointing Devices

Device	Working	Use
Mouse	Detects movement with an optical sensor or laser; has left, right buttons and scroll wheel	General navigation
Trackball	Fixed device with a rolling ball moved by the thumb or fingers	Limited desk space
Touchpad	Touch-sensitive surface that tracks finger movement	Laptops

Device	Working	Use
Joystick	Lever whose tilt direction controls movement	Games, simulators
Light pen	Light-sensitive pen that detects screen position	Drawing on CRT screens (old)
Touchscreen	Detects touch via capacitive or resistive layers	Phones, tablets, kiosks, ATMs
Graphics tablet and stylus	Pen draws on a pressure-sensitive pad	Digital art, design

3. Scanning and Image Input

- **Scanner:** captures paper documents or photos as digital images. Types include flatbed, sheet-fed, handheld and drum scanners.
- **Digital camera:** captures photos and videos
- **Webcam:** captures live video for calls and streaming

4. Recognition Devices (Automatic Data Entry)

Device	Full Form	What It Reads	Use
OCR	Optical Character Recognition	Printed or typed text from images	Digitizing books, documents
OMR	Optical Mark Recognition	Pencil marks or filled bubbles	Exam answer sheets, surveys
MICR	Magnetic Ink Character Recognition	Magnetic ink characters	Bank cheques
BCR	Barcode Reader	Barcode (black and white lines)	Supermarkets, libraries
QR scanner	Quick Response	2D QR code	Payments, tickets
RFID reader	Radio Frequency Identification	Data from RFID tags	Access cards, toll tags, inventory
Smart card reader	n/a	Chip cards	ID, banking

5. Biometric Input Devices

They identify a person using physical traits.

- **Fingerprint scanner**
- **Face recognition camera**

- **Iris/retina scanner**
- **Voice recognition system**

6. Audio Input

- **Microphone:** converts sound waves into electrical signals for recording, calls and voice commands
- **MIDI keyboard:** inputs musical notes

7. Other Input Devices

- **Sensors:** temperature, motion, GPS and other physical sensors (used in smart devices)
- **Remote control and gamepad**
- **Motion-capture and VR controllers**

Output Devices

Definition

An output device converts the computer's processed digital data into a form humans can understand: text, images, video, sound or physical objects.

Characteristics

- Only receives data from the computer (one-way)
- Presents information as **soft copy** (screen, sound) or **hard copy** (paper)

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1. Visual Display Devices (Soft Copy)

Monitor (VDU: Visual Display Unit)

Type	Technology	Notes
CRT	Cathode Ray Tube fires electron beams at a phosphor screen	Heavy, bulky, high power use (obsolete)
LCD	Liquid crystals controlled by electric current, lit by a backlight	Flat, light, low power
LED	LCD monitor with LED backlighting	Brighter, more efficient
OLED	Each pixel produces its own light	True blacks, thin, flexible screens
Touchscreen monitor	Display with touch layer	Input and output both

Monitor specifications

- **Screen size:** measured diagonally in inches

- **Resolution:** number of pixels (HD 1280×720, Full HD 1920×1080, 4K 3840×2160)
- **Refresh rate:** how many times per second the image updates, in Hz (60 Hz, 144 Hz)
- **Response time:** how quickly pixels change color, in ms
- **Aspect ratio:** width to height (16:9, 21:9)
- **Panel type:** IPS (best colors), VA (good contrast), TN (fast, cheaper)

Projector

Casts the display onto a wall or screen. Types: LCD, DLP and LED/laser projectors. Used in classrooms, meetings and cinemas.

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2. Printers (Hard Copy)

Impact Printers

Strike an ink ribbon against paper.

Printer	Description	Use
Dot matrix	Pins form characters from dots; noisy, prints carbon copies	Invoices, bills
Daisy wheel	Fully formed characters on a wheel (obsolete)	Typewriter-quality text
Line printer	Prints a whole line at once; very fast	Large reports on mainframes

Non-Impact Printers

No physical strike; quiet and better quality.

Printer	Working	Pros and Cons
Inkjet	Sprays microscopic ink droplets	Great color and photos; ink can be costly
Laser	Laser draws an image on a drum; toner is fused with heat	Fast, sharp text, low cost per page; higher upfront price
Thermal	Heat on special paper or ribbon	Receipts, labels; fades over time
3D printer	Builds objects layer by layer from plastic or resin	Prototyping, medical, manufacturing

Printer measurements

- **DPI:** dots per inch (print quality)
- **PPM:** pages per minute (speed)
- **CPS:** characters per second (impact printers)

Plotter

Draws large, high-precision drawings using pens or inkjet heads. Used for engineering, architecture and maps (CAD output).

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3. Audio Output

- **Speakers:** convert electrical signals into sound
- **Headphones/earphones:** private audio
- **Sound card:** converts digital audio to analog signals for these devices

4. Other Output Devices

- **Braille displays:** raised dots for blind users
- **Haptic devices:** vibration feedback (game controllers, phones)
- **Robotic and actuator outputs:** control machines
- **VR/AR headsets:** immersive display and sound

Devices That Do Both (Input/Output)

Device	Input Role	Output Role
Touchscreen	Detects touch	Displays image
Modem / NIC	Receives data	Sends data
Headset	Microphone	Speakers
Multifunction printer (MFP)	Scans	Prints
External drive / pen drive	Provides data	Stores data
Smartphone / tablet	Touch, camera, mic	Screen, speaker

Input vs Output Devices

Feature	Input Device	Output Device
Purpose	Sends data to the computer	Shows results to the user
Direction	User → Computer	Computer → User

Feature	Input Device	Output Device
Data conversion	Human form → Digital	Digital → Human form
Examples	Keyboard, mouse, scanner, mic	Monitor, printer, speaker, projector

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Soft Copy vs Hard Copy

Soft Copy	Hard Copy
Temporary display on screen or sound	Permanent printout on paper
Cannot be held	Can be held and shared physically
Examples: monitor, speaker	Examples: printer, plotter

Ports Used to Connect Them

Port	Common Devices
USB	Keyboard, mouse, printer, scanner, webcam
PS/2	Older keyboards and mice
HDMI / DisplayPort / VGA / DVI	Monitors, projectors
3.5 mm audio jacks	Speakers, headphones, microphone
Bluetooth / Wi-Fi	Wireless peripherals